

Question 1 (10 Marks)	START A NEW PAGE	Marks
(a) Find the gradient of the tangent to the curve $y = e^{\sin x}$ at $x = \frac{\pi}{3}$.		2
(b) Differentiate with respect to x .		
(i) $\cot 3x$.		2
(ii) $\ln\left(\frac{x}{x^2 + 1}\right)$.		2
(c) Express $0.4\dot{6}$ as an geometric series and hence write its equivalent fraction in simplest terms.		2
(d) Show that $y = e^{\frac{1}{x}}$ has no stationary points. Justify your answer.		2

Question 2 (10 Marks) **START A NEW PAGE**

(a) Evaluate $\int_1^2 \frac{x^2 - 9}{x^3} dx$		2
(b) Find		
(i) $\int \tan^2(2x - 1) dx$.		2
(ii) $\int \frac{3x}{x - 1} dx$.		2
(c) A weather balloon released at ground level 2100m from an observer rises at a rate of 100 metres per minute.		
(i) Show that, after t minutes, the height of the balloon, h metres, is given by $h = 2100 \tan \theta$, where θ is the angle of elevation.		1
(ii) Find the rate, in radians per minute, at which the angle of elevation of the observer is increasing when the balloon is at an altitude of 1400 metres.		3

Question 3 (10 Marks)**START A NEW PAGE****Marks**

- (a) The n th term of a series is given by $T_n = n + \left(\frac{1}{3}\right)^n$. 4

Find the sum of the first 10 terms.

- (b) Prove, by mathematical induction for $n = 1, 2, 3, \dots$ and $x > 0$, 4

$$\frac{1}{x(x+1)^n} = \frac{1}{x} - \frac{1}{x+1} + \frac{1}{(x+1)^2} - \dots + \frac{1}{(x+1)^n}.$$

- (c) Given $f(x) = 2^{3x}$, find $f'(0)$. 2

Question 4 (10 Marks)**START A NEW PAGE**

- (a) (i) The inner surface of a bowl is formed by rotating the curve $y = \ln(x - 2)$ about the y -axis between $y = 0$ and $y = 2$. 3
Calculate the volume of water that this bowl holds when the depth of water is filled to a depth of h units.

- (ii) If water is poured into the bowl at a rate of 50 cubic units per second, find the *exact* rate at which the water level is rising when the depth of water is 1.5 units. 2

- (b) (i) Express $\sqrt{3} \cos 2x - \sin 2x$ in the form of $A \cos(2x + \alpha)$ 2
for $0 < \alpha < \frac{\pi}{2}$ and $A > 0$.

- (ii) Hence, or otherwise, sketch the curve $y = \sqrt{3} \cos 2x - \sin 2x$; $0 \leq x \leq 2\pi$. 2

- (iii) Find the first positive solution for $\sqrt{3} \cos 2x - \sin 2x = 1$. 1

Question 5 (10 Marks)**START A NEW PAGE****Marks**

- (a) (a) Given that $\sum_{k=1}^n \frac{k}{(k+1)!} = 1 - \frac{1}{(n+1)!}$, show that **3**

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} + \frac{n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!}$$

- (b) Sketch the graph of $y = \ln[2x(3-x)]$, showing all important features. **4**

- (c) Find $\int \frac{x^2}{\sqrt{9-x^2}} dx$ using the substitution $\alpha = \sin^{-1}\left(\frac{x}{3}\right)$. **3**

Question 6 (10 Marks)**START A NEW PAGE**

- (a) Calculate the sum of the series **4**

$$2^{5x} + 2^{3x} + 2^x + \dots + 2^{(11-6k)x}.$$

- (b) Amy borrowed \$280 000 to purchase an apartment at a reducible interest rate of 7.8 % per annum compounded monthly from 2006 from her mortgage lender. She makes equal monthly repayments of \$ M until the loan is repaid.

- (i) How much did Amy owe after her first monthly repayment of \$ M ? **1**

- (ii) Calculate Amy's equal monthly instalment \$ M , if her loan is for 25 years. **2**

- (iii) After 1 year, the Reserve Bank increases the interest rate by 0.25% *pa*, however, her mortgage lender increases her interest rate by a further 0.1% *pa*. How much extra does Amy need to pay, per month, with the total extra interest rate rise? **3**

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Question 1 (10 Marks)**START A NEW PAGE****Marks**

(a) $\frac{dy}{dx} = \cos x e^{\sin x}$ when $x = \frac{\pi}{3}$, $\frac{dy}{dx} = \frac{1}{2} e^{\frac{\sqrt{3}}{2}}$

2

(b) (i) $\frac{d}{dx} \cot(3x) = -1(\tan(3x))^{-2} \times 3\sec^2(3x)$

2

$$= -\frac{3\sec^2 3x}{\tan^2 3x} = -3 \operatorname{cosec}^2 3x$$

(ii) $\frac{d}{dx} \ln\left(\frac{x}{x^2+1}\right) = \frac{d}{dx} (\ln x - \ln(x^2+1)) = \frac{1}{x} - \frac{2x}{x^2+1}$

$$= \frac{-x^2+1}{x(x^2+1)}$$

2

(c) $0.4\dot{6} = 0.4 + 0.06666... = 0.4 + 0.06 + 0.006 + 0.0006 + ...$

2

$$A = 0.06, r = 0.1 \rightarrow S_{\infty} = \frac{0.06}{1-0.1} = \frac{1}{15}$$

$$\therefore 0.4\dot{6} = \frac{4}{10} + \frac{1}{15} = \frac{7}{15}$$

(d) $y' = -\frac{e^{\frac{1}{x}}}{x^2}$. Now $e^{\frac{1}{x}} \neq 0$ as $x \neq 0$.

2

Question 2 (10 Marks)**START A NEW PAGE**

(a) $\int_1^2 \frac{1}{x} - 9x^{-3} dx = \left[\ln x + \frac{9}{2x^2} \right]_1^2 = \ln 2 - \frac{27}{8}$

2

(b) $\int \tan^2(2x-1) dx$

2

$$= \int [\sec^2(2x-1) - 1] dx$$

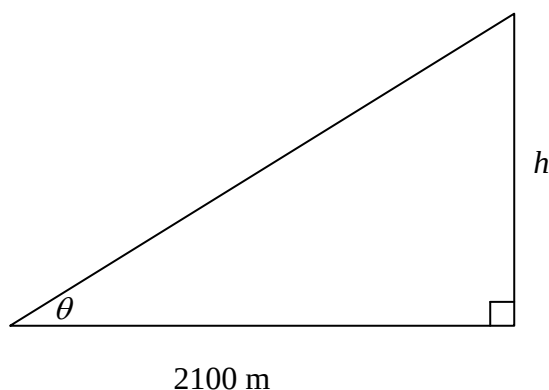
$$= \frac{1}{2} \tan(2x-1) - x + c$$

(ii) $\int \frac{3x}{x-1} dx$

2

$$= \int \frac{3(x-1)+3}{x-1} dx = \int 3 + \frac{3}{x-1} dx = 3x + 3\ln(x-1) + c$$

(c)



$$\frac{dh}{dt} = 100$$

4 mks

$$\text{And } \frac{h}{2100} = \tan \theta$$

$$\therefore h = 2100 \tan \theta$$

$$\frac{dh}{d\theta} = 2100 \sec^2 \theta$$

$$\text{Now } \frac{d\theta}{dt} = \frac{d\theta}{dh} \cdot \frac{dh}{dt}$$

$$\therefore \frac{d\theta}{dt} = \frac{\cos^2 \theta}{2100} \times 100$$

$$\text{But when } h = 1400, \tan \theta = \frac{2}{3} \rightarrow \cos^2 \theta = \frac{9}{13}$$

$$\therefore \frac{d\theta}{dt} = \frac{9}{21 \times 13} = \frac{3}{91} \text{ radian per minute.}$$

Question 3 (10 Marks)

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$$(a) T_1 = 1 + 1/3, T_2 = 2 + 1/9, T_3 = 3 + 1/27, \dots$$

$\therefore$ we have an AP + GP sum.

$$\therefore \text{AP sum} = 1 + 2 + 3 + \dots + 10 = 55$$

$$\text{GP sum} = 1/3 + 1/9 + 1/27 + \dots + 1/3^{10}$$

$$= \frac{\frac{1}{3} \left(1 - \left(\frac{1}{3} \right)^{10} \right)}{\frac{2}{3}} = \frac{1}{2} \left(1 - \frac{1}{3^{10}} \right)$$

$$\therefore \text{total sum} = 55 + \frac{1}{2} \left(1 - \frac{1}{3^{10}} \right)$$

(b) **Test $n = 1$,**

$$\begin{aligned} \text{LHS} &= \frac{1}{x(x+1)^1} & \text{RHS} &= \frac{1}{x} - \frac{1}{x+1} \\ & & &= \frac{x+1-x}{x(x+1)^1} \\ & & &= \frac{1}{x(x+1)} \end{aligned}$$

$\therefore$ true for $n = 1, x > 0$

Assume true for $n = k$

$$\frac{1}{x(x+1)^k} = \frac{1}{x} - \frac{1}{x+1} - \frac{1}{(x+1)^2} - \dots - \frac{1}{(x+1)^k}$$

For $n=k+1$

$$\begin{aligned} & \frac{1}{x} - \frac{1}{x+1} - \frac{1}{(x+1)^2} - \dots - \frac{1}{(x+1)^k} - \frac{1}{(x+1)^{k+1}} \\ &= \frac{1}{x(x+1)^k} - \frac{1}{(x+1)^{k+1}} \quad \text{using the assumption} \\ &= \frac{x+1-x}{x(x+1)^{k+1}} \\ &= \frac{1}{x(x+1)^{k+1}} \end{aligned}$$

$\therefore$ Since true for $n=1$ and proven true for $n=k+1$, assuming true for $n=k$
It is also true for all $n \geq 1$.

(c) $f(x) = 2^{3x} = 8^x$
 $f'(x) = \ln 8 \cdot 8^x$
 $\therefore f'(0) = \ln 8$

2

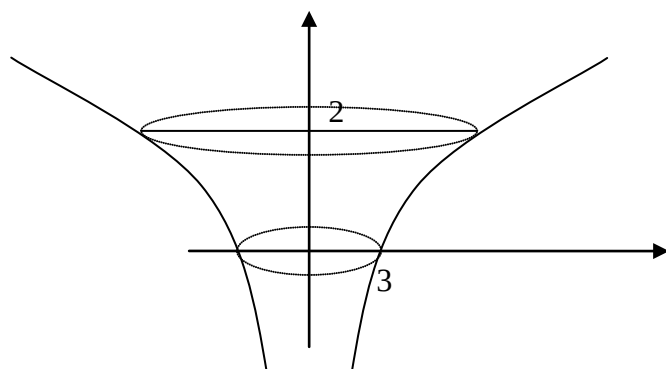
Question 4 (10 Marks)

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Marks

(a) (i)

3



$$\begin{aligned} \text{Vol} &= \pi \int_0^h (2 + e^y)^2 dy = \pi \left[4 + 4e^y + \frac{1}{2}e^{2y} \right]_0^h = \pi \left[4h + 4e^h + \frac{1}{2}e^{2h} - 4 - \frac{1}{2} \right] \\ &= \pi \left[4h + 4e^h + \frac{1}{2}e^{2h} - \frac{9}{2} \right] u^3. \end{aligned}$$

(ii) $\frac{dV}{dt} = 50, \quad \frac{dV}{dh} = \pi(4 + 4e^h + e^{2h})$

2

$$\therefore 50 = \pi(4 + 4e^{1.5} + e^3) \cdot \frac{dh}{dt} \quad \therefore \frac{dh}{dt} = \frac{50}{\pi(4 + 4e^{1.5} + e^3)} \text{ units/sec.}$$

(b) (i) $\sqrt{3} \cos 2x - \sin 2x \equiv A[\cos 2x \cos \alpha - \sin 2x \sin \alpha]$

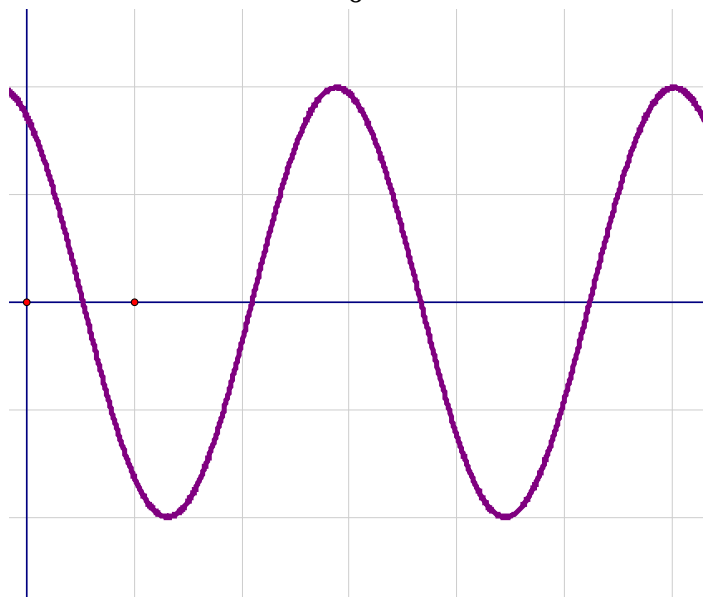
$$\therefore A = \sqrt{3+1} = 2, \quad \cos \alpha = \sqrt{3}, \quad \sin \alpha = 1 \quad \therefore \tan \alpha = \frac{1}{\sqrt{3}} \quad \therefore \alpha = \frac{\pi}{6}$$

2

$$\therefore \sqrt{3} \cos 2x - \sin 2x = 2 \cos(2x + \frac{\pi}{6})$$

$$(ii) \text{ Let } y = 2 \cos(2x + \frac{\pi}{6})$$

2



$$(iii) \quad 2 \cos(2x + \frac{\pi}{6}) = 1$$

1

$$\cos(2x + \frac{\pi}{6}) = \frac{1}{2}$$

$$\therefore 2x + \frac{\pi}{6} = \frac{\pi}{3} \quad \therefore 2x = \frac{\pi}{6} \quad \therefore x = \frac{\pi}{12}$$

Question 5 (10 Marks)

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$$\begin{aligned} (a) \quad \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} + \frac{n+1}{(n+2)!} &= \sum_{k=1}^n \frac{k}{(k+1)!} + \frac{n+1}{(n+2)!} \\ &= 1 - \frac{1}{(n+1)!} + \frac{n+1}{(n+2)!} \\ &= 1 - \frac{n+2}{(n+2)!} + \frac{n+1}{(n+2)!} \\ &= 1 + \frac{-n-2+n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!} \end{aligned}$$

3

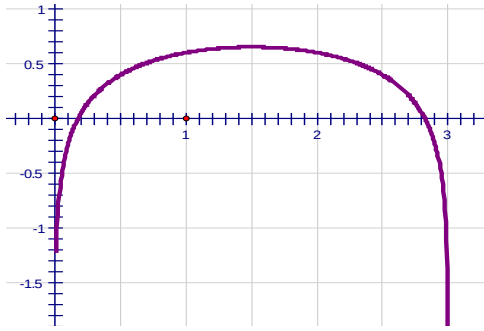
$$(b) \quad y = \ln 2x + \ln(3-x); \quad x \neq 0, 3$$

4

$$y' = \frac{1}{x} + \frac{-1}{3-x} = \frac{3-2x}{x(3-x)} \quad \therefore \text{tp at } x = 1.5, y = \ln(4.5)$$

Test tp

$$y'' = \frac{-1}{x^2} + \frac{-1}{(3-x)^2} = -\frac{4}{9} - \frac{4}{9} = -\frac{8}{9} < 0 \quad \therefore \text{absolute max since only 1 tp.}$$



NOTE: Curve is asymptotic at $x = 0$ and $x = 3$.

(c) $x = 3\sin\alpha \therefore dx = 3\cos\alpha d\alpha$

3

$$\begin{aligned} \therefore \int \frac{9\sin^2\alpha \cdot 3\cos\alpha}{\sqrt{9-9\sin^2\alpha}} d\alpha &= \int \frac{27\sin^2\alpha \cos\alpha}{3\cos\alpha} d\alpha = \int 9\left(\frac{1-\cos 2\alpha}{2}\right) d\alpha \\ &= \frac{9}{2}\left(\alpha - \frac{1}{2}\sin 2\alpha\right) + c \end{aligned}$$

Question 6 (10 Marks)

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(a) Let $S = 2^{5x} + 2^{3x} + 2^x + \dots + 2^{(11-6k)x}$ and as a GS $r = 2^{-2x}$

4

So $2^{-2x} \cdot S = 2^{3x} + 2^{5x} + 2^{7x} + \dots + 2^{(11-6k)x} + 2^{(9-6k)x}$.

$S(1 - 2^{-2x}) = 2^{5x} - 2^{(9-6k)x}$.

$$\therefore S = \frac{2^{5x} - 2^{(9-6k)x}}{1 - 2^{-2x}} = \frac{2^{7x} - 2^{(11-6k)x}}{2^{2x} - 1}.$$

Note the number of terms was $N = 3k-2$.

(b) (i) $R = \frac{13}{2000}$ per month. (0.0065)

1

$280\,000 \times 1.0065 - M$

(ii) Month 1: $280\,000 \times 1.0065 - M$

2

Month 2: $280\,000 \times 1.0065^2 - M(1 + 1.0065)$

⋮
⋮
⋮

Month 300: $280\,000 \times 1.0065^{300} - M(1 + 1.0065 + 1.0065^2 + \dots + 1.0065^{299})$

$$\begin{aligned} \therefore M &= \frac{280\,000 \times 1.0065^{300}}{1(1.0065^{300} - 1) \div 0.0065} \\ &= \text{\$2 124.12} \end{aligned}$$

(iii) Amount remaining after 1 year = $280\,000 \times 1.0065^{12} - 2124.12 \times \frac{(1.0065^{12} - 1)}{0.0065}$.

$$= \text{\$276 217.22}$$

3

New interest rate = 8.15%

$$M_1 = \frac{276\,217.22 \times \frac{24163}{24000}}{\left(\frac{24163}{24000} - 1\right) \div \frac{163}{24000}} = \$2187.37$$

$\therefore$ extra payments with the extra 0.35% is $\$2187.37 - \$2\,124.12 = \text{\$63.25}$